

Entropy (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Predict whether a physical or chemical change will result in an increase or decrease in entropy of a system.
- Calculate standard entropy changes, $\Delta S^\ominus$, from standard entropy values, $S^\ominus$.

In our daily lives, we encounter many other instances of increased disorder. Think of an ice cube melting: the particles that were once organised in the cube, are now dispersing as a liquid. Perhaps you have heard about statues made of calcium carbonate, CaCO_3 , that are slowly degrading due to exposure to acid rain.



Credit: Shenki, [Getty Images](https://www.gettyimages.com/detail/photo/ice-cube-royalty-free-image/104789662)

(<https://www.gettyimages.com/detail/photo/ice-cube-royalty-free-image/104789662>)



Credit: Photography by Jason Gallant, [Getty Images](#)

(<https://www.gettyimages.com/detail/photo/statue-at-angkor-thom-north-gate-royalty-free-image/969274980>)

Figure 1. Water disperses when ice melts and calcium carbonate statues dissolve when exposed to acid rain.

Have you noticed that in all the examples described previously, the tendency is for an increase in disorder? Why is that?

Entropy

In all these examples there is an increase in entropy.

Entropy (S) can be thought of as a measure of disorder or randomness in a system. It specifically refers to the distribution of energy among the particles in a system. Consequently, this means that in a system with higher entropy, the particles are more spread out and moving more. Imagine a closed jar where you placed coloured sand in beautifully organised layers. The smallest movement of the jar will cause the grains of sand to move and mix. The more the jar is moved, the more the colours will mix and they will never return to their original arrangement, no matter how many times you shake the jar. The system tends to remain in randomness as this means the energy is more dispersed.

Recall from [section S1.1.1a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) when you learned about solids, liquids and gases and how the particles were arranged. Which state of matter seems the most ordered to you?

In **Interactive 1** below, order the three systems according to an increasing level of entropy (or disorder).



✓ Check

Interactive 1. Increasing entropy.

Nature of Science

Aspect: Patterns and trends

The laws of thermodynamics have changed in content and numbering through history, combining the ideas of chemists and physicists to make sense of temperature, energy and entropy.

The first and second laws of thermodynamics were developed by various scientists in the 19th century. The third law is more recent and was formulated by a physical chemist called Walter Nernst in 1906. It states that as the temperature of a pure substance reaches zero kelvin (absolute zero), its entropy will also reach a minimum. The increasing understanding of the way matter behaves at extremely low temperatures continues to lead to various scientific discoveries in the field of low-temperature physics, such as the discovery of superconductivity and superfluidity.

Interactive 2 shows the relative entropy of solids, liquids, and gases. Under the same conditions, solids (with their fixed arrangements of particles) have the lowest entropy, and gases (with their random arrangements of particles) have the highest

entropy. In gases, energy can be distributed in more ways than in liquids and solids – they have higher entropy.

Interactive 2. Entropy and the States of Matter.

 More information for interactive 2

An interactive slide illustrates the relative entropy of solids, liquids, and gases. The structure of the states of matter is as follows:

Solid: Particles are tightly packed in a fixed, orderly arrangement.

Liquid: Particles are close together but not in a fixed arrangement.

Gases: Particles are loosely packed and spread out randomly.

Entropy increases from solid to liquid and then to gas, as indicated by a rightward arrow at the top.

Entropy decreases from gas to liquid and then to solid, as indicated by a leftward arrow at the bottom.

This interactive slide helps users understand the relationship between the states of matter and entropy and the particle arrangement in solids, liquids, and gases.

International Mindedness

One of the major issues that we face on the planet currently is sustainability. Various countries around the world found the need to create protected areas both on land and on the ocean to preserve and protect ecosystems. Marine protected areas, for example, are designated areas in the ocean where human activities are regulated, promoting sustainable fisheries, protecting vulnerable marine areas, and conserving biodiversity.

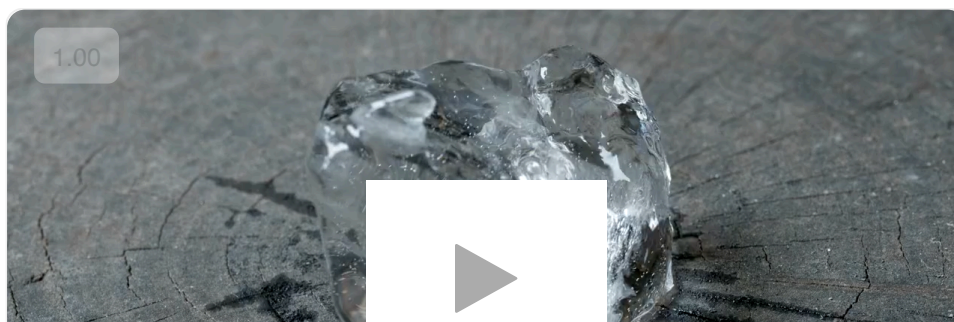
In order to have sustainable systems the focus is on maintaining their order and structure over time. You might now be pausing to consider how your understanding of the concept of entropy can impact your perspective on this issue.

As you learned, all systems tend to increase in entropy, which is the tendency for systems to become disordered and unstable. In order to promote sustainability worldwide, it is required an understanding of the importance of sustainability and the need for global cooperation to address environmental issues and promote sustainable development. The study of entropy can help us appreciate the challenges involved in creating sustainable global systems and the need for coordinated efforts to address these challenges.

Changes in entropy (ΔS) for physical changes

Recall from [section S1.1.1a \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) that the state of a substance depends on its temperature. Solid states exist at lower temperatures and gas states at higher temperatures. If the temperature of a substance changes, its state might also change.

During changes of state, there is a change in the entropy of a system. When an ice cube melts, the water molecules that were compacted neatly in a crystalline shape start moving past each other. This represents an increase in entropy as the kinetic energy of the particles increases.



Video 1. An Ice Cube Melting Is Undergoing a Change in Entropy.

Credit: dogholiday, [Getty Images](https://www.gettyimages.com/detail/video/melting-ice-cube-time-lapse-stock-footage/1064566246)

(<https://www.gettyimages.com/detail/video/melting-ice-cube-time-lapse-stock-footage/1064566246>)

 More information for video 1

An interactive video demonstrates the melting process of an ice cube placed on a wooden surface. The ice cube is transparent but contains a few air bubbles trapped inside.

As the ice gradually melts, it transforms into liquid water. The water spreads across the wooden surface of the wood, seeping into the natural cracks of the wood.

A certain amount of water content on the outside slowly evaporates, leaving behind moisture traces.

This interactive helps users understand the physical changes of water, including melting and evaporation.

The play button with a slider is provided at the bottom of the video for the user to play and pause the video as required. The speedometer, next to the slider, with playback rates 0.25, 0.5, 1, 1.25, 1.5, and 2 allows the user to adjust the speed of the video. The sound icon, next to this, allows users to play the video with sound or in mute. The zoom-out at the bottom right enables the video to be played in full-screen mode

In the reverse process, when water freezes the particles that were able to move past each other in the liquid water get trapped into a crystalline structure and ice forms. In this case, there was an increase in order, which means a decrease in entropy.

Figure 2 shows a summary of the changes of state and the corresponding changes of entropy.

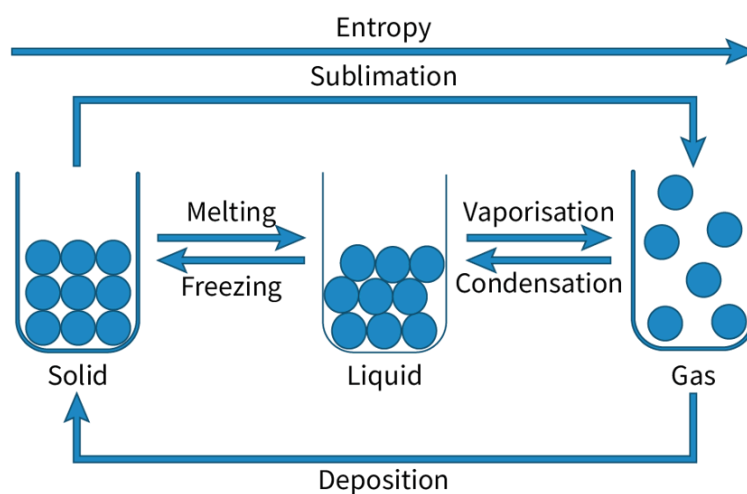


Figure 2. Changes of state and corresponding entropy.

 More information for figure 2

The diagram illustrates the changes in states of matter and the corresponding changes in entropy. It highlights three main states: solid, liquid, and gas, each represented by circles in different containers.

- **Solid to Liquid:** The diagram shows an arrow labeled 'Melting' pointing from solid to liquid and an arrow labeled 'Freezing' in the opposite direction, indicating the reversible process between these states.
- **Liquid to Gas:** Similarly, arrows labeled 'Vaporization' and 'Condensation' illustrate the reversible transition between liquid and gas states.
- **Solid to Gas:** The process of 'Sublimation' is represented by an arrow pointing directly from solid to gas.
- **Gas to Solid:** The arrow labeled 'Deposition' shows the direct transition of gas to solid.

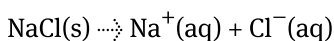
At the top of the diagram, there is an overarching arrow labeled 'Entropy,' indicating that transition processes affect entropy levels, with gas assumed to have the highest entropy compared to solid and liquid based on common scientific understanding.

[Generated by AI]

Entropy change (ΔS) can be negative ($-\Delta S$) or positive ($+\Delta S$), depending on whether there is a decrease in entropy or an increase in entropy. When considering the sign of the entropy change (positive or negative), it is important to consider the changes in the state of the reactants and products. For example, evaporation results in an increase in entropy ($+\Delta S$) as gases have higher entropy than liquids. In chemical reactions that involve solids or liquids being converted into gaseous products, there will be a large increase in entropy.



Recall the example with layers of coloured sand in a jar. Think of each colour of sand as a pure substance. After the container is shaken, you have made a mixture. The combination of pure substances to form homogeneous or heterogeneous mixtures, also leads to an increase in entropy. For example, when you place sodium chloride (solute) into water (solvent), the solid salt dissolves, which means its ions become dispersed between the water molecules – you have created a solution. This represents an increase in the entropy of the system, $+\Delta S$, as shown in **Figure 3**.



This process can also be represented by the equation:

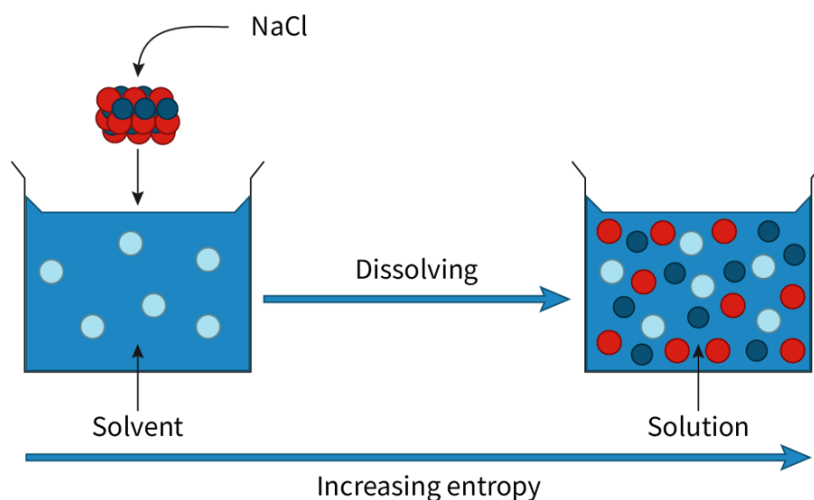
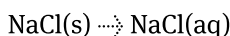


Figure 3. Entropy of a dissolution.

 More information for figure 3

The image is a diagram illustrating the process of dissolving NaCl (sodium chloride) in water and the resulting increase in entropy. On the left, there's a container labeled 'Solvent' containing water with a few light blue circles representing water molecules. Above, there is a cluster of red and dark blue circles labeled 'NaCl' representing the solid salt, with an arrow indicating it is being introduced into the water. In the center, the word 'Dissolving' is written with an arrow pointing to the right.

To the right, a second container labeled 'Solution' contains a mix of red, light blue, and dark blue circles, indicating that the salt has dissolved in the water, creating a homogenous solution. Below the diagram, a horizontal arrow

labeled 'Increasing entropy' signifies the increase in entropy during the dissolution process.

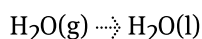
[Generated by AI]

For the reverse process, if the solvent is evaporated and the salt crystallised there is a decrease in the entropy of the system, $-\Delta S$.

Use the interactive to classify the changes of entropy (increase/decrease) in the examples of processes that result in a change in entropy.

Worked example 1

Examine the equation below that shows the condensation of water.

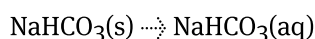


Predict if there is an increase or decrease in entropy and whether the sign for the change in entropy is positive or negative.

As there is a change from gas to liquid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 2

Examine the equation for dissolving sodium hydrogen carbonate to form a solution.



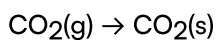
Predict if there is an increase or decrease in entropy and the sign for the change in entropy.

As there is a change from solid to a solution. There is an increase in entropy, so ΔS is positive.

Worked example 3

Write the equation, including state symbols, for the deposition of carbon dioxide to create dry ice.

Predict if there is an increase or decrease in entropy and the sign for the change in entropy.

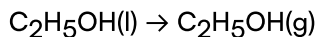


As there is a change from gas to solid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 4

Write the equation, including state symbols, for the vaporisation of ethanol, $\text{C}_2\text{H}_5\text{OH}$.

Predict if there is an increase or decrease in entropy and the sign for the change in entropy.



As there is a change from liquid to gas, there is an increase in entropy. ΔS is therefore positive.

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance.
- Demonstrate how to initiate and plan a CAS experience.

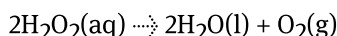
The melting of ice is an example of a physical process that results in an increase in entropy. Can you think of a global environmental issue that involves the melting of ice?

According to [Climate.gov](https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level#:~:text=Global%20mean%20sea%20level%20has,of%20seawater%20as%20it%20warms.)  (<https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level#:~:text=Global%20mean%20sea%20level%20has,of%20seawater%20as%20it%20warms.>) the global sea level has risen by approximately 20 cm since 1880. This is caused by the melting of the polar ice caps. For example, according to [NASA](https://climate.nasa.gov/vital-signs/ice-sheets/#:~:text=Antarctica%20is%20losing%20ice%20mass,adding%20to%20sea%20level%20C)  (<https://climate.nasa.gov/vital-signs/ice-sheets/#:~:text=Antarctica%20is%20losing%20ice%20mass,adding%20to%20sea%20level%20C>) Antarctica is losing 1.4×10^{14} kg of ice each year.

A possible CAS experience is to research the causes and effects of the changing sea levels and raise awareness of this issue in your school or the local community.

Changes in entropy (ΔS) in chemical reactions

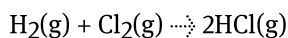
In the examples above, only physical changes were considered. However, there can also be a change in entropy in chemical reactions. An example is the decomposition of hydrogen peroxide:



From the balanced equation, we can see that two moles of solution in the reactants become two moles of liquid plus one mole of gas in the products, which represents an increase in entropy. A change in the amount (in mol) of gaseous reactants or

products can be used to predict a change in entropy. An increase in the number of moles of gas creates an increase in entropy and a decrease in the number of moles of gas on the products side signifies a decrease in entropy.

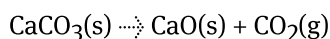
It is also important to note that if there are the same number of moles of gas in the reactants and products, then the entropy change will be quite small. Here is an example of a synthesis reaction of hydrochloric acid:



From the balanced equation, we can see that two moles of gas in the reactants become two moles of gas in the products, which represents a minimal and unknown change in entropy.

Worked example 5

Examine the equation below for the decomposition of calcium carbonate.

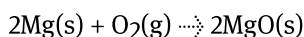


Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

There is one mole of gas in the products and only solids in the reactants.
There is an increase in entropy; ΔS is therefore positive.

Worked example 6

Examine the equation below for the synthesis of magnesium oxide.



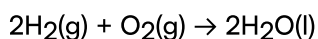
Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

There is one mole of gas in the reactants and only solids in the product.
As there is a change from gas to solid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 7

Write the equation, including state symbols, for the formation of water from oxygen gas and hydrogen gas.

Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

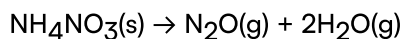


There are three moles of gas in the reactants and only liquid in the product. As there is a change from gas to liquid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 8

Write the equation, including state symbols, for the decomposition of ammonium nitrate powder, NH_4NO_3 , into nitrous oxide gas (N_2O) and water vapour.

Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.



There are three moles of gas in the products and only solids in the reactants. As there is a change from solid to gas, there is an increase in entropy. ΔS is therefore positive.

Standard entropy values ($S^\ominus$) and changes in entropy ($\Delta S^\ominus$)

The standard entropy of a substance is the entropy change from heating the substance from absolute zero (0 K) to the standard temperature of 298 K. A perfect crystal at absolute zero (0 K) has a standard entropy of zero. Therefore, all substances have a positive standard entropy value.

Making connections

Subtopic S1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-40864/>).—Why is the entropy of a perfect crystal at 0 K predicted to be zero?

As you learned in subtopic S1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-40864/>), the lower the temperature of a substance, the more its particles tend to occupy a fixed position in a lattice structure, where they vibrate in place. The more order exists in that crystalline structure, the lower the entropy, as you have been learning in this section.

At absolute zero (0 K), all atomic motion stops, and the system is in a state of perfect order, with all atoms occupying specific positions in a regular lattice structure. Entropy is a measure of the number of possible arrangements or configurations of the system's particles. At absolute zero, the system is in its lowest energy state, and there is only one possible arrangement of particles that corresponds to that energy state. Therefore, the number of possible arrangements is zero, and the entropy is also zero.

In practice, it is not possible to cool a crystal to absolute zero (-273.15°C), but it has been confirmed experimentally that as the temperature approaches zero kelvin, the entropy approaches zero. In October 2021, scientists broke the record for the coldest temperature ever measured in a lab. They achieved the bone-chilling temperature of 38 trillionths of a degree above -273.15°C by dropping magnetised gas 120 metres (393 feet) down a tower.

Standard entropy values can be used to calculate the standard entropy change ($\Delta S^\ominus$) for a reaction, using the following equation:

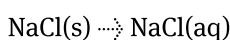
$$\Delta S^\ominus = \Sigma S^\ominus (\text{products}) - \Sigma S^\ominus (\text{reactants})$$

Study skills

Standard entropy values ($S^\ominus$) can be found in the DP chemistry data booklet in [section 13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>). Note that its units are $\text{J K}^{-1} \text{mol}^{-1}$.

Worked example 9

Consider the dissolution of sodium chloride in water.



Standard entropy values:

$$S^\ominus (\text{NaCl(s)}) = +72.1 \text{ J K}^{-1} \text{mol}^{-1}$$

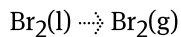
$$S^\ominus (\text{NaCl(aq)}) = +115.5 \text{ J K}^{-1} \text{mol}^{-1}$$

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$S^\ominus (\text{NaCl(s)}) = +72.1 \text{ J K}^{-1} \text{mol}^{-1}$ $S^\ominus (\text{NaCl(aq)}) = +115.5 \text{ J K}^{-1} \text{mol}^{-1}$
Step 2: Write out the equation.	$\Delta S^\ominus = \Sigma S^\ominus (\text{products}) - \Sigma S^\ominus (\text{reactants})$
Step 3: Substitute the values given.	$= 115.5 - 72.1$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding	$= +43.4 \text{ J K}^{-1} \text{mol}^{-1}$ Note that the positive sign shows solute particles separate and dis

Worked example 10

Consider the physical change of bromine from liquid to gas.



Standard entropy values:

$$S^\ominus (\text{Br}_2(\text{l})) = +152 \text{ J K}^{-1} \text{ mol}^{-1}$$

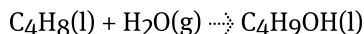
$$S^\ominus (\text{Br}_2(\text{g})) = +245.5 \text{ J K}^{-1} \text{ mol}^{-1}$$

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$S^\ominus (\text{Br}_2(\text{l})) = +152 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{Br}_2(\text{g})) = +245.5 \text{ J K}^{-1} \text{ mol}^{-1}$
Step 2: Write out the equation.	$\Delta S^\ominus = \Sigma S^\ominus (\text{products}) - \Sigma S^\ominus (\text{reactants})$
Step 3: Substitute the values given.	$= 245.5 - 152$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= +93.5 \text{ J K}^{-1} \text{ mol}^{-1}$ Note that the positive sign shows bromine moves from the liquid phase to the gas phase.

Worked example 11

Consider the reaction of but-1-ene (C_4H_8) with steam to form butanol ($\text{C}_4\text{H}_9\text{OH}$):



Standard entropy values:

$$S^\ominus (\text{C}_4\text{H}_8(\text{l})) = +310 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{H}_2\text{O}(\text{g})) = +189 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{C}_4\text{H}_9\text{OH}(\text{l})) = +228 \text{ J K}^{-1} \text{ mol}^{-1}$$

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the	$S^\ominus (\text{C}_4\text{H}_8(\text{l})) = +310 \text{ J K}^{-1} \text{ mol}^{-1}$

Solution steps	Calculations
question.	$S^\ominus (\text{H}_2\text{O}(\text{g})) = +189 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{C}_4\text{H}_9\text{OH}(\text{l})) = +228 \text{ J K}^{-1} \text{ mol}^{-1}$
Step 2: Write out the equation.	$\Delta S^\ominus = \Sigma S^\ominus (\text{products}) - \Sigma S^\ominus (\text{reactants})$
Step 3: Substitute the values given.	$= 228 - (310 + 189)$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= -271 \text{ J K}^{-1} \text{ mol}^{-1}$ Note that the negative sign shows that the reaction is not spontaneous.

Worked example 12

Consider the combustion of propene and the following standard entropy values:



Standard entropy values:

$$S^\ominus (\text{C}_3\text{H}_6(\text{l})) = +267 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{O}_2(\text{g})) = +205 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{H}_2\text{O}(\text{g})) = +189 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{CO}_2(\text{g})) = +214 \text{ J K}^{-1} \text{ mol}^{-1}$$

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$S^\ominus (\text{C}_3\text{H}_6(\text{l})) = +267 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{O}_2(\text{g})) = +205 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{H}_2\text{O}(\text{g})) = +189 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{CO}_2(\text{g})) = +214 \text{ J K}^{-1} \text{ mol}^{-1}$

Solution steps	Calculations
Step 2: Write out the equation.	$\Delta S^\ominus = \Sigma S^\ominus (\text{products}) - \Sigma S^\ominus (\text{reactants})$
Step 3: Substitute the values given.	$= 3(+214) + 3(+189) - [(+267) + 3(+214)]$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding	$= +19.5 \text{ J K}^{-1} \text{ mol}^{-1}$ Note that the positive sign shows a spontaneous process. 1 mole of liquid and 4.5 moles of gas are produced.

Complete the summary table in **Interactive 3** by dragging the listed reactions to the correct space.

ΔS negative (entropy decreases)	ΔS positive (entropy increases)
Changes from gas $\rightarrow$ liquid $\rightarrow$ solid	Changes from solid $\rightarrow$ liquid $\rightarrow$ gas
Freezing	Melting
Condensation	Vaporisation
Deposition	Sublimation
Crystallisation	Dissolution
Decreasing number of moles of gas	Increasing number of moles of gas

H₂O(s) $\rightarrow$ H₂O(l)

2H₂(g) + O₂(g) $\rightarrow$ 2H₂O(l)

CO₂(s) $\rightarrow$ CO₂(g)

CaCO₃(s) $\rightarrow$ CaO(s) + CO₂(g)

H₂O(l) $\rightarrow$ H₂O(g)

NaCl(aq) $\rightarrow$ NaCl(s)

H₂O(g) $\rightarrow$ H₂O(l)

H₂O(l) $\rightarrow$ H₂O(s)

NaCl(s) $\rightarrow$ NaCl(aq)

I₂(g) $\rightarrow$ I₂(s)

☒ Check

Interactive 3. Complete the Summary Table.

Theory of Knowledge

Entropy is the measurement of disorder, which is constantly increasing in the universe.

What knowledge is there to be gained by measuring something which is always predictably increasing?

Watch **Video 2** for a visual representation of a model entropy system in a car.

The Statistical Interpretation of Entropy



Video 2. Entropy explained through a model car.

Now watch **Video 3** to hear the conclusion that can be made from the model.

The Statistical Interpretation of Entropy



Video 3. Conclusions from modelling entropy.

Complete the activity below to practise the content studied in this section.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/whole class activity

Create your own model for entropy:

- Select one example from everyday life where there is a high or low entropy (e.g. a messy room or the ingredients before a meal is prepared).
- Take a photo of your original system and explain why it has high or low entropy.
- Produce a change in entropy in your system (e.g. clean up your room, or prepare a meal with the ingredients selected).
- Take a photo of your new system and explain why its entropy has increased or decreased when compared to your original system.
- Present your work in class.

Gibbs energy and spontaneous changes (HL)

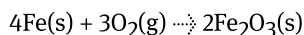
Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

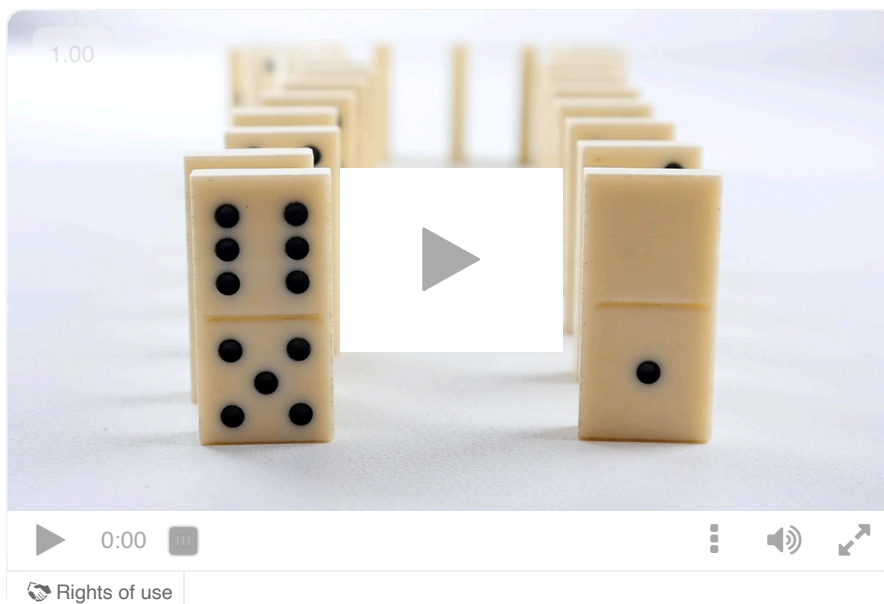
- (R1.4.2) Apply the equation $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$ to calculate unknown values of these terms.
- (R1.4.3) Interpret the sign of ΔG calculated from thermodynamic data.
- (R1.4.3) Determine the temperature at which a reaction becomes spontaneous.

In our daily lives, we encounter many reactions that happen without the need of consistent energy input. For example, when you strike a match to paper, the paper will continue burning even after you blow the match. If an iron nail is exposed to air rusts (iron(III) oxide) forms, according to the equation:



However, iron(III) oxide (rust) will not break down to form iron and oxygen, nor can you return the paper to its original state once it has burned. The burning of paper and the rusting of a nail are known as spontaneous reactions. Their inverse reactions are called non-spontaneous reactions.

Spontaneous reactions will happen once the energy needed to start the reaction is provided. Like the touch that the finger gives to the first domino in **Video 1**:



Video 1. Visualisation of the Activation Energy Needed to Start a Spontaneous Reaction.

Credit: magicmore, Getty Images

(<https://www.gettyimages.com/detail/video/white-dominoes-falling-in-chain-reaction-domino-effect-stock-footage/853285342>)

 More information for video 1

An interactive video features a series of vertically arranged dominoes forming an almost inverted U-shape. As the video progresses, a person's finger gradually pushes the first domino on the left, which has eleven dots, causing it to fall and knock over the next domino. The dominoes keep toppling one by one until the last one on the right, with one dot, falls.

This falling domino effect represents a spontaneous reaction, where an initial push sets off a continuous process without requiring additional input.

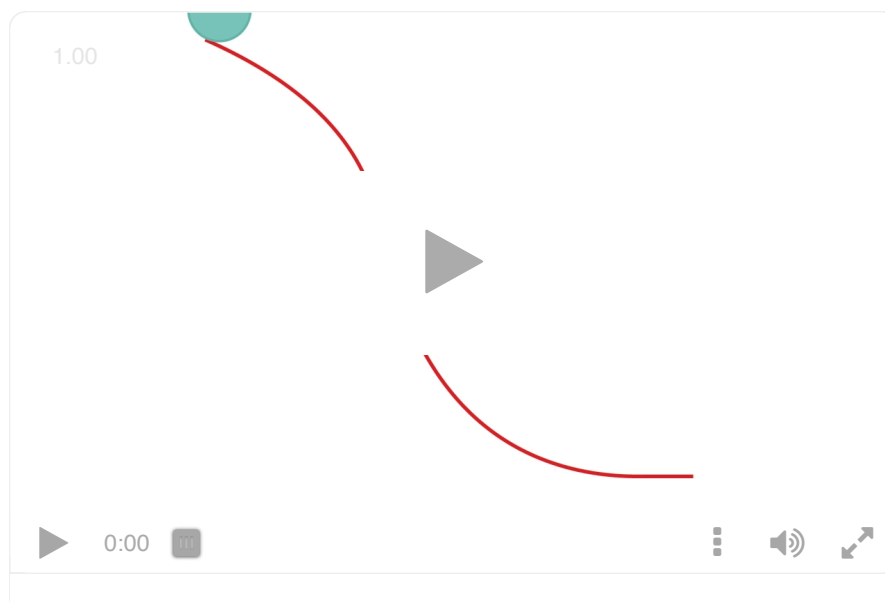
How can we determine if a reaction will happen spontaneously?

What factors determine the feasibility of a reaction?

Spontaneous reactions

A spontaneous reaction occurs without the addition of energy, other than the energy required to start the reaction (also called activation energy). For example, if you strike a match to paper, the paper will continue to burn by itself, even if you remove the match – this is a spontaneous reaction. But if you want to fry potato chips, you need to continuously provide heat until they are cooked – this is a non-spontaneous reaction.

A spontaneous process can be compared to a ball rolling down a hill; once the ball is given the initial push, it will continue to roll down the hill until it reaches the bottom (**Interactive 1**). A non-spontaneous reaction is one that needs a constant input of energy to occur. It can be compared to a ball rolling up a hill, which only occurs if energy is constantly supplied to get the ball to the top.



Interactive 1. A Spontaneous and a Non-spontaneous Process.

 More information for interactive 1

The interactive video is shown with a blue ball placed at the top of a curved red hill. The interactive compares spontaneous and non-spontaneous processes.

Spontaneous Process: A ball rolls down a downward-sloping curve smoothly, representing a process that occurs naturally without external input.

Non-spontaneous Process: The same ball moves uphill along the same curve, indicating that an external force or energy is required for the process to occur.

This interactive helps users understand the difference between spontaneous and non-spontaneous processes by illustrating energy changes through ball movement.

Making connections

Activation energy is the energy required to start a chemical reaction. You will learn more about it in [section R2.2.4](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>).

The second law of thermodynamics (**Figure 1**) states that the total entropy of the universe tends to increase. Simply put, in a spontaneous process, the energy goes from a concentrated form to a dispersed form (entropy increases). For a spontaneous process, the total entropy of the system and surroundings (ΔS_{total}) must increase. This can be represented in equation form as:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

Certain chemical reactions have a decrease in the entropy of the system but are still spontaneous because the entropy of the surroundings increases to a much greater extent, which gives an overall increase in entropy for the process. One example is the formation of crystals in a salt solution when the solvent evaporates. The crystals (system) have a lower entropy (more ordered state) than the salt solution, but the solvent molecules increase in entropy as they are lost to the surrounding as a gas (more disordered state).

The same happens with our planet as part of the universe. It was mentioned in the last section that the universe is expanding, which means it is becoming more disordered – its entropy is increasing. However, on Earth we can find many examples of ordered systems, like the rocks that make up most of the planet. More order means that the entropy of the system is lower. Nonetheless, we need to consider that overall entropy is still increasing, as the surroundings of our planet (the universe) are increasing in entropy faster than entropy decreases in our planet (the system).

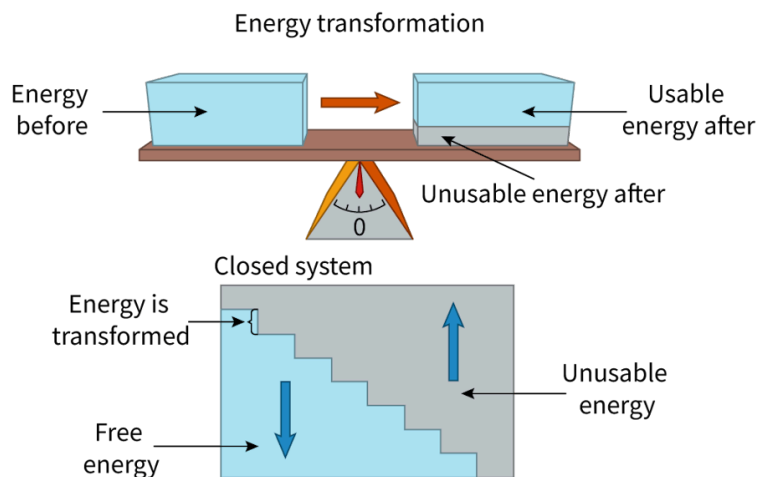


Figure 1. The second law of thermodynamics.

[More information for figure 1](#)

The diagram illustrates concepts related to thermodynamics. At the top, there is a seesaw-like setup with two rectangular blocks of ice placed on it, moving left and right. Arrows indicate the directions of movement, with an orange arrow showing the general direction from left to right. At the fulcrum, there is a triangular indicator set at zero, suggesting equilibrium. Below, there is a gradient, step-like schematic with arrows. A small arrow pointing to the left is shown on the left side, and larger arrows show movement downward and upward. Components are labeled with descriptive text explaining different parts of the process, potentially relating to heat exchange, equilibrium, and energy transformation processes in thermodynamics.

[Generated by AI]

Gibbs energy (G)

The increase in the entropy of the surroundings is usually in the form of heat released from the system into the surroundings. You can feel it if you place your hand close to a fire.

For the surroundings, the entropy change is given by the relationship in **Table 1**.

Table 1. Equation for entropy change in the surrc

Equation	Symbols
$\Delta S_{\text{surroundings}} = -\Delta H/T$	$\Delta S_{\text{surroundings}}$ = the entropy change
	ΔH = the enthalpy change of the process
	T represents the absolute temperature

Notice the change of sign: the heat given out by the system (ΔH for the system will be negative) enters the surroundings, so any decrease in the enthalpy of the system equates to an addition of heat to the surroundings ($\Delta S_{\text{surroundings}}$ will be positive). This equation can be transformed to the one in **Table 2**.

Table 2. Equation for the change in standard Gibbs energy under standard conditions temperature and pressure.

Equation	Symbols	Unit
$\Delta G^{\ominus} = \Delta H^{\ominus} - T \Delta S^{\ominus}$	$\Delta G^{\ominus}$ = the change in standard Gibbs energy under standard conditions.	kilojoule mol ⁻¹ (kJ mol ⁻¹)
	$\Delta H^{\ominus}$ = the standard enthalpy under standard conditions.	kilojoule mol ⁻¹ (kJ mol ⁻¹)
	T = temperature	kelvin
	$\Delta S^{\ominus}$ = the standard entropy change	joules per mole per kelvin (J K ⁻¹ mol ⁻¹)

A change in Gibbs energy, ΔG , takes into account the direct entropy change resulting from the transformation of the chemicals and the indirect entropy change of the surroundings resulting from the transfer of heat energy.

Study skills

Thermodynamic data values are given in the DP chemistry data booklet in **section 13** (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>).

When calculating ΔG , carefully check the units of ΔH (which are generally kJ mol⁻¹) and ΔS (which are generally J K⁻¹ mol⁻¹) and convert them if necessary. All temperatures must be in kelvin.

Generally, you will want to convert joules to kilojoules, which means you need to divide the number by 1000.

Nature of Science

Aspect: Models

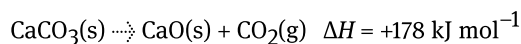
For DP chemistry, in the context of studying entropy, models can be used to predict whether the entropy change for a given reaction or process is positive or negative. For example, the calculation of the standard entropy change of a reaction uses a model that considers the absolute entropy

values of each substance involved in the reaction and the number of moles of each substance to calculate the total absolute entropy on each side of the reaction. The difference between the total absolute entropy on the left-hand side and the right-hand side gives us the change in entropy for the reaction. This model allows us to predict whether the entropy change for a given reaction or process is positive or negative.

It is now clear that there are three factors to be considered when determining the spontaneity of a reaction:

- the sign of ΔH
- the sign of ΔS
- the temperature at which the reaction takes place.

This allows us to explain the occurrence of endothermic reactions, such as the decomposition of calcium carbonate to form calcium oxide and carbon dioxide:



Work through these two examples to learn why endothermic reactions can be spontaneous.

Worked example 1

Calculate the ΔG for the decomposition of calcium carbonate at 298 K knowing that ΔH for the reaction is $+178 \text{ kJ mol}^{-1}$ and the ΔS is $+161 \text{ J K}^{-1} \text{ mol}^{-1}$.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$\Delta H = +178 \text{ kJ mol}^{-1}$ $\Delta S = +161 \text{ J K}^{-1} \text{ mol}^{-1}$ $T = 298 \text{ K}$
Step 2: Convert any values that need changing.	ΔS need dividing by 1000 $\Delta S = \frac{+161}{1000}$ $= +0.161 \text{ kJ K}^{-1} \text{ mol}^{-1}$
Step 3: Write out the equation.	$\Delta G = \Delta H - T\Delta S$
Step 4: Substitute the values given.	$= +178 - 298 \times +0.161$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= +130 \text{ kJ mol}^{-1}$

Worked example 2

Calculate the ΔG for the decomposition of calcium carbonate at 1106 K knowing that ΔH for the reaction is $+178 \text{ kJ mol}^{-1}$ and the ΔS is $+161 \text{ J K}^{-1} \text{ mol}^{-1}$.

Solution steps	Calculati
Step 1: Write out the values given in the question.	$\Delta H = +$ $\Delta S = +$ $T = 1106$
Step 2: Convert any values that need changing.	ΔS need dividing b $\Delta S = \frac{+}{1}$ $= +$
Step 3: Write out the equation.	$\Delta G = \Delta$
Step 4: Substitute the values given.	$= (+178$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= -0.06$

As you can see from these calculations, Gibbs energy change (ΔG) can be either positive or negative. For any reaction, the sign of ΔG is dependent only on the temperature at which the reaction is happening, as the enthalpy and entropy are the same for the same reaction. A reaction is spontaneous when ΔG is negative.

Fill in the blank in **Interactive 2** to complete the statement.

Fill in the missing number

So for the decomposition of calcium carbonate, the reaction is spontaneous when the temperature is K or above.

☒ Check

Interactive 2. Fill in the Blank.

The spontaneity of a reaction is dependent on the sign of the Gibbs energy change (ΔG). This is the ultimate deciding factor that tells us whether a reaction is spontaneous or not, as follows:

- If the Gibbs energy change, ΔG , is negative, the reaction is spontaneous.
- If the Gibbs energy change, ΔG , is positive, the reaction is non-spontaneous.
- If the Gibbs energy change, ΔG , is zero, the reaction is at equilibrium.

Using the equation $\Delta G = \Delta H - T\Delta S$, complete **Interactive 3**, which summarises the factors affecting the spontaneity of a reaction.

	$\Delta H < 0$	$\Delta H > 0$
$\Delta S > 0$		
$\Delta S < 0$		

Non-spontaneous at all T ($\Delta G > 0$)	Spontaneous at low T (when $T\Delta S$ is small)	Spontaneous at all T ($\Delta G < 0$)	Spontaneous at high T (when $T\Delta S$ is large)
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✓ Check

Interactive 3. Complete the table.

🔗 More information for interactive 3

An interactive table with drop boxes represents the factors affecting the spontaneity of a reaction. The table contains two columns labeled $\Delta H < 0$ and $\Delta H > 0$, and two rows labeled $\Delta S > 0$ and $\Delta S < 0$. Each row or column has two dotted drop boxes.

Users need to drag and drop the descriptions of spontaneity into the appropriate drop boxes in the table.

The boxed drag-and-drop options provided at the bottom from left to right are:

Non-spontaneous at all T ($\Delta G > 0$)

Spontaneous at low T (when $T\Delta S$ is small)

Spontaneous at all T ($\Delta G < 0$)

Spontaneous at high T (when $T\Delta S$ is large)

This interactivity helps users understand how enthalpy (ΔH) and entropy (ΔS) influence the Gibbs free energy (ΔG) and the spontaneity of chemical

reactions.

The “Check” icon at the bottom left allows users to check their answers. The right answer is indicated with a slider. If the inputted answers are wrong, the “Retry” icon at the bottom left allows users to retry the activity.

Read below for the solution.

Column 1 Row 1: Spontaneous at all T ($\Delta G < 0$)

Column 2 Row 1: Spontaneous at high T (when $T\Delta S$ is large)

Column 1 Row 2: Spontaneous at low T (when $T\Delta S$ is small)

Column 2 Row 2: Non-spontaneous at all T ($\Delta G > 0$)

Looking at **Interactive 3**, you can see that there are two cases in which the value of the ΔG is independent of the temperature:

- Exothermic reactions that result in an increase in entropy — the value of the ΔG will be negative at any temperature:

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (-) - (+)(+) \\ &= (-) \quad \text{always spontaneous}\end{aligned}$$

- Endothermic reactions with a decrease in entropy — the value of the ΔG is positive at any temperature.

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (+) - (+)(-) \\ &= (+) \quad \text{never spontaneous}\end{aligned}$$

Temperature is a factor in the other two scenarios:

- Exothermic reactions with a decrease in entropy are only spontaneous at low temperatures.

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (-) - (+)(-) \\ &= (-) \text{ if } T \text{ is low (spontaneous) OR} \\ &\quad (+) \text{ if } T \text{ is high (non-spontaneous)}\end{aligned}$$

- Endothermic reactions with an increase in entropy are only spontaneous at high temperatures. If the temperature is high enough, subtracting the product of $T\Delta S$ from the ΔH value will give a negative (spontaneous) ΔG value. At lower temperatures, the product of $T\Delta S$ is not large enough to cancel out the positive ΔH value, so the ΔG value is positive (non-spontaneous).

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (+) - (+)(+) \\ &= (-) \text{ if } T \text{ is high (spontaneous) OR} \\ &\quad (+) \text{ if } T \text{ is low (non-spontaneous)}\end{aligned}$$

Making connections

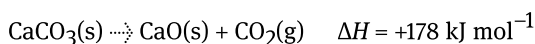
Section R1.3.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>) — Cells need to have a spontaneous reaction to provide electricity. What does it mean with respect to Gibbs energy?

You might notice that batteries you buy from the store are labelled with the voltage (1.5 V, 6 V, 9 V, etc). This measurement of the energy a cell can provide is known as the cell potential. The cell potential is a positive value and you have learned in this section that for the reaction to be spontaneous, Gibbs energy must be a negative value. What is the relationship between the cell potential and the Gibbs energy for an electrochemical cell?

Making connections

In section R3.2.14 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/14%20gibbs-free-energy-and-standard-electrode-potentials-hl-id-44784/>), you will learn more about cells and calculating their potential. How can electrochemical data also be used to predict the spontaneity of a reaction?

You have already seen the equation and enthalpy for the decomposition of calcium carbonate:



Apply your knowledge to this reaction to answer the following questions.

Worked example 3

State whether the decomposition of calcium carbonate is endothermic or exothermic.

Endothermic, as the enthalpy change is positive.

Worked example 4

What can you deduce about the entropy for the reaction?

Entropy is positive as there is more disorder in the products than reactants due to one mole of solid being converted into one mole of solid and one mole of gas.

Worked example 5

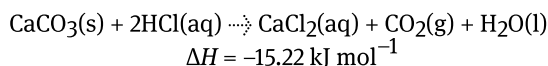
What temperature conditions are likely to lead to a spontaneous reaction. Justify your answer.

$$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$$

The reaction will become spontaneous at elevated temperatures as $-T\Delta S^\ominus$ will become greater than $\Delta H^\ominus$ which will lead to a negative $\Delta G^\ominus$.

At lower temperatures, $\Delta G^\ominus \approx \Delta H^\ominus$, which means Gibbs energy will be positive and the reaction will be non-spontaneous.

Consider the reaction of calcium carbonate with hydrochloric acid and answer the following questions.



Worked example 6

Predict, stating a reason, the sign of $\Delta S^\ominus$ for the above reaction.

The sign of $\Delta S^\ominus$ is positive because one mole of solid plus two moles of aqueous reactants produce one mole of aqueous plus one mole of gaseous plus one mole of liquid products, so there is an increase in entropy.

Worked example 7

Calculate the standard entropy change for the reaction.

Note:

$$S^\ominus (\text{CaCO}_3(\text{s})) = +38.1 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{HCl}(\text{aq})) = +53.1 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{CaCl}_2(\text{aq})) = +124 \text{ J K}^{-1} \text{ mol}^{-1}$$

Use section 13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet.

Solution steps	Calculations
Step 1: Write out the values given in the question and obtain any other values you need from the DP chemistry data booklet.	$S^\ominus (\text{CaCO}_3(\text{s})) = +38.1 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{HCl}(\text{aq})) = +53.1 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{CaCl}_2(\text{aq})) = +124 \text{ J K}^{-1} \text{ mol}^{-1}$ From the DP chemistry data booklet $S^\ominus (\text{CO}_2(\text{g})) = +214 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{H}_2\text{O}(\text{l})) = +70 \text{ J K}^{-1} \text{ mol}^{-1}$
Step 2: Write out the equation.	$\Delta S^\ominus = \Delta S^\ominus (\text{products}) - \Delta S^\ominus (\text{reactants})$
Step 3: Substitute the values given.	$= [(+124) + (+214) + (+70)] - [$

Solution steps	Calculations
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$\Delta S^{\ominus} = +264 \text{ J K}^{-1} \text{ mol}^{-1}$ (3 s.f.)

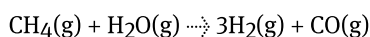
Worked example 8

Determine the value of $\Delta G^{\ominus}$ for the reaction at 298 K and explain if the reaction is spontaneous at this temperature.

Solution steps	Calculations
Step 1: Use the values previously presented /calculated.	$\Delta H^{\ominus} = -15.22 \text{ kJ mol}^{-1}$ $\Delta S^{\ominus} = +264 \text{ J K}^{-1} \text{ mol}^{-1}$
Step 2: Convert any values that need changing.	$\Delta S^{\ominus} = +0.264 \text{ kJ K}^{-1} \text{ mol}^{-1}$
Step 3: Write out the equation.	$\Delta G^{\ominus} = \Delta H^{\ominus} - T \Delta S^{\ominus}$
Step 4: Substitute the values given.	$= (-15.22) - (298)(0.264)$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= -93.9 \text{ kJ (3 s.f.)}$ Given that $\Delta G^{\ominus}$ is negative, the reaction is spontaneous at this temperature.

Worked example 9

The reaction between methane and steam is non-spontaneous at 298 K.



Determine the temperature above which the reaction becomes spontaneous given the following data:

$$\Delta H = 205 \text{ kJ mol}^{-1}$$

$$\Delta S = 216 \text{ J K}^{-1} \text{ mol}^{-1}$$

Solution steps	Calculations
Step 1: Use the information provided in the question.	For the reaction to be spontaneous you calculate the temperature at which $\Delta G^\ominus = 0$ to tell at which temperature the reaction becomes spontaneous.
Step 2: Convert any values that need changing.	$\Delta S^\ominus = 0.216 \text{ kJ K}^{-1} \text{ mol}^{-1}$
Step 3: Write out the equation.	$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$
Step 4: Substitute the values given.	$0 = (205) - T(0.216)$ $-205 = -0.216T$ $T = \frac{-205}{-0.216}$ $= 949$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 949 \text{ K (3 s.f.)}$ This reaction becomes spontaneous above 949 K (or 1222 °C).

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Consider the boiling of water.

- Write a balanced chemical equation to represent the change of state of water.
- Use section 13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of your DP chemistry data booklet to calculate the enthalpy of this process.

Explain why the boiling point of pure water is 373 K.

- Use your knowledge of intermolecular forces to choose another substance that has a lower boiling point than water. Show calculations to support your choice.

Gibbs energy and equilibrium (HL)

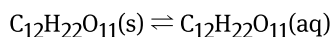
Higher level (HL)

Learning outcomes

By the end of this section you should be able to perform calculations using the equation $\Delta G = \Delta G^\ominus + RT \ln Q$ and its application to a system at equilibrium $\Delta G^\ominus = -RT \ln K$.

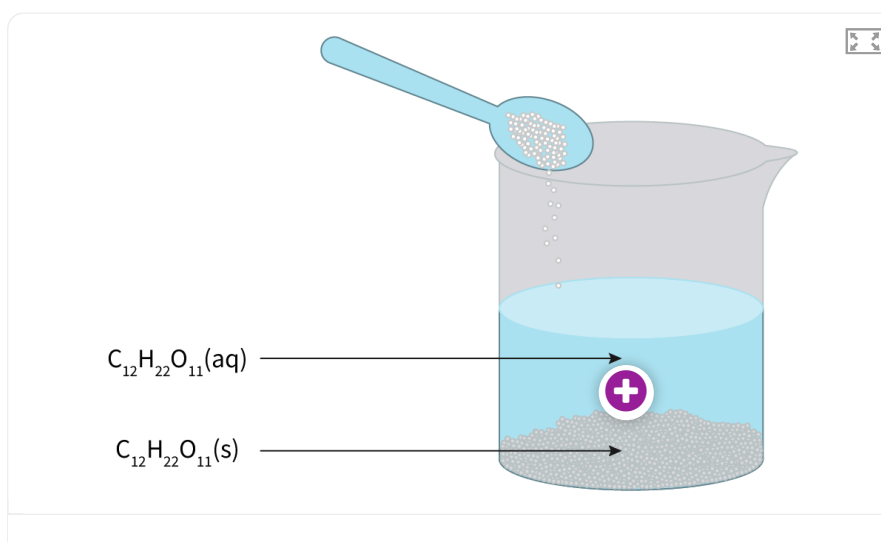
So far, you have learned about various reactions where reactants are transformed into products in unidirectional reactions. For example, when a log of wood burns, the compounds that made up the wood will be irreversibly changed into carbon dioxide and water.

However, there are processes and reactions that can exist with a balance between reactants and products. Think of a time when you have added sugar to your tea. Maybe you noticed that if you add a lot of it, no matter how long you stir, you will not be able to dissolve more of your sugar and you end up with a little pile at the bottom of your cup. What happens is that the system reaches an equilibrium, where the amount of sugar that is dissolving, is the same as the amount of sugar returning to the solid state. This system is said to be in equilibrium and it can be represented by the following equation:



Notice that there is a double-headed half-arrow representing the equilibrium of the process of dissolution of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, which is the compound present in white sugar. What do you think the double-headed arrow implies for this process?

Use **Interactive 1** to investigate how sugar dissolves in tea.



Interactive 1. Solution of Sugar in Tea at Equilibrium.

 More information for interactive 1

An interactive image illustrates the process of dissolving sugar in water. A spoonful of solid sugar, $C_{12}H_{22}O_{11}(s)$, is added to a beaker containing water. As the sugar dissolves, it forms an aqueous solution, $(C_{12}H_{22}O_{11}(aq))$, represented by dispersed molecules in the liquid. The image highlights the distinction between undissolved solid sugar at the bottom and dissolved sugar in the solution. The hotspot, when clicked, pops up another image which can be described as follows:

A magnified view of the dissolving process shows individual sugar molecules separating from the solid and dispersing into the solution. Arrows indicate movement, representing the dynamic equilibrium between dissolved molecules and undissolved sugar.

Given what you know now about spontaneity of reactions, you might be wondering:

What happens to ΔG for systems that reach an equilibrium?

Reversible reactions and equilibrium

So far, we have considered that chemical reactions transform reactants into products in a one-way direction. However, there are some reactions that are reversible. That means that the products of the forward reaction can be used as reactants to yield the original compounds (backward reaction.) When that happens we say the system is in equilibrium.

For example, bromine, Br_2 , is a liquid at room temperature, but it is very volatile. This means that it can easily turn into a gas. Bromine has a brown colour as you can see in **Video 1**.

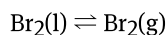
Bromine is scary



Video 1. Bromine, Br_2 , being poured and the gas evolving.

When bromine is initially placed in a closed container, some of the particles move from the liquid phase to the gas phase. This is expected, as you know that entropy of the system should increase, which means that the ΔG is negative. You can see the brown liquid at the bottom and a brown gas starting to fill the container.

But after a while, there are no more visible changes to the unaided eye. Once the gas phase has been filled with as many bromine particles as possible in the space available, the number of particles moving in both directions (liquid-to-gas phase and gas-to-liquid phase) is the same. This can be represented as:



The forward reaction is $\text{Br}_2(\text{l}) \rightarrow \text{Br}_2(\text{g})$, and the backward reaction is $\text{Br}_2(\text{g}) \rightarrow \text{Br}_2(\text{l})$.

Notice again the double-headed arrow ($\rightleftharpoons$) showing that the system is at equilibrium and be careful not to confuse it with the double headed arrow for resonance structures ($\leftrightarrow$).

Figure 1 shows a visual representation of what is happening inside a closed jar with bromine.

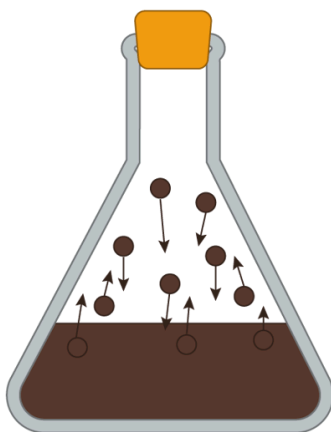


Figure 1. Bromine, Br_2 , at equilibrium: the rate of evaporation is equal to the rate of condensation.

 More information for figure 1

The image depicts a laboratory flask with a stopper at the top, containing bromine depicted as a brown liquid at the bottom. Within the flask, there are arrows pointing upwards from the liquid, representing the evaporation of bromine molecules into the gas phase, and arrows pointing downward, indicating condensation back into liquid form. This illustrates the dynamic equilibrium where the rate of evaporation equals the rate of condensation.

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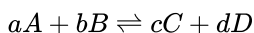
Making connections

You can read more about systems in equilibrium in [subtopic R2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>).

For reversible reactions, we can determine the extent of the reaction by calculating the equilibrium constant, K , and the reaction quotient Q .

The equilibrium constant is calculated using equilibrium concentrations of reactants and products.

The reaction quotient is calculated using the concentrations of reactants and products when the system is not at equilibrium. For example, for the general reaction:



the equilibrium constant is written as:

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

where a , b , c and d represent the stoichiometric coefficients in the balanced equation. The reaction quotient (Q) is written the same way, but the concentrations of products and reactants used should be the ones when the system is not at equilibrium.

When writing these expressions, there are a few important notes:

- the products are on the top of the expression and the reactants on the bottom
- square brackets [] are used to show the concentrations (in mol dm^{-3}) of the reactants and products
- the powers to which the concentrations are raised are the stoichiometric coefficients in the balanced equation for the reaction.

Study skills

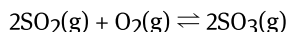
Note that the equilibrium constant is temperature dependent. This means that it has different values at different temperatures.

Also note that the magnitude of K is an indication of the extent to which a chemical reaction favours the formation of products or reactants at equilibrium.

- When $K \gg 1$ this means that at equilibrium, there are relatively more products than reactants, indicating that the reaction strongly favours the formation of products.
- When $K \ll 1$, it means that at equilibrium, there are relatively more reactants than products, indicating that the reaction strongly favours the formation of reactants.

Worked example 1

Write an expression for the equilibrium constant, K , for the reaction of sulfur dioxide (SO_2) with oxygen (O_2) to form sulfur trioxide (SO_3):



$$K = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]}$$

Note that when $Q = K$, the reaction is at equilibrium (**Figure 2**).

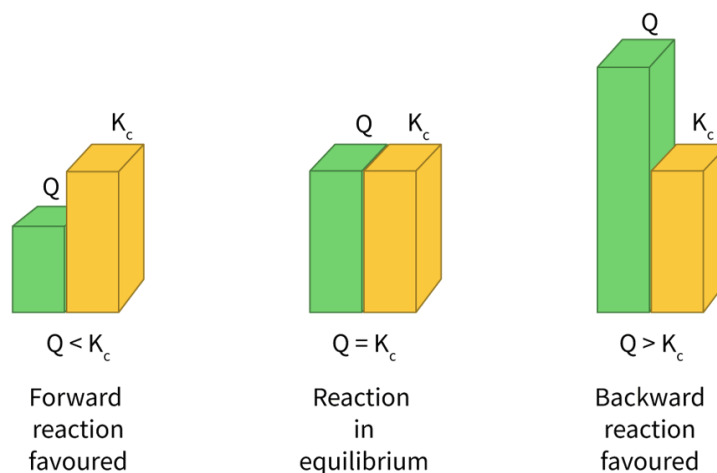


Figure 2. The importance of the relative sizes of Q and K in determining the direction of the reaction to reach equilibrium.

More information for figure 2

The image is a bar chart illustrating the comparative sizes of Q and K , which are variables in a chemical reaction to determine the direction needed to reach equilibrium. The chart consists of three sets of bars. In the first set, the green bar representing Q is shorter than the yellow bar representing K . In the second set, the green bar representing Q and the yellow bar representing K are of equal height, indicating equilibrium. In the third set, the green bar representing Q is taller than the yellow bar representing K . The Y-axis is labeled as "Size."

[Generated by AI]

Gibbs energy for reactions at equilibrium

The relationship between the Gibbs energy change (ΔG) and the spontaneity of a chemical reaction was covered in section [R1.4.2-3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/>). Recall that, for a spontaneous reaction, the Gibbs energy change must be negative.

We can use the following equation in **Table 1** to calculate the Gibbs energy change for a reaction not at equilibrium (note the use of the reaction quotient, Q).

Table 1. Equation for Gibbs energy change.

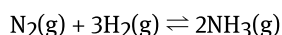
Equation	Symbols	Units
$\Delta G = \Delta G^\ominus + RT \ln Q$	ΔG = the Gibbs energy change	kilojoules per mole (kJ mol^{-1})
	$\Delta G^\ominus$ is the standard Gibbs energy change for the reaction	kilojoules per mole (kJ mol^{-1})
	R is the universal gas constant ($8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)	joules per kelvin mole ($\text{J K}^{-1} \text{ mol}^{-1}$)

Equation	Symbols	Units
	T = the absolute temperature	kelvin (K)
	$\ln Q$ = the natural logarithm of the reaction quotient	no units

Note that R has the units $\text{J K}^{-1} \text{mol}^{-1}$. To convert to kJ mol^{-1} , divide by 1000.

Worked example 2

Consider the reaction to produce ammonia.



The standard Gibbs energy, $\Delta G^\ominus$, is $-32.7 \text{ kJ mol}^{-1}$ and the reaction quotient, Q , is 6.4×10^{-7} .

Determine the change in Gibbs energy, ΔG , for this reaction when $T = 100^\circ\text{C}$.

Solution steps	Calculations
Step 1: Write out the information provided in the question.	$\Delta G^\ominus = -32.7 \text{ kJ mol}^{-1}$ $Q = 6.4 \times 10^{-7}$
Step 2: Convert any values that need changing.	$T = 100 + 273$ $= 373 \text{ K}$
Step 3: Write out the equation.	$\Delta G = \Delta G^\ominus + RT \ln Q$
Step 4 Substitute the values given.	$= -32.7 + \left(\frac{8.31}{1000} \right) (373) \ln(6.4 \times 10^{-7}) =$ Note that the gas constant, R , has been divided $\text{K}^{-1} \text{mol}^{-1}$.
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$\Delta G = -77 \text{ kJ mol}^{-1}$ (2 s.f.) Given that ΔG is negative, this means the reaction is favoured at $T = 100^\circ\text{C}$ (which will move the system towards equilibrium).

The standard Gibbs energy change, $\Delta G^\ominus$, for a reversible reaction can also be determined for systems at equilibrium taking into consideration the equilibrium constant, K , using the equation in **Table 2**.

Table 2. Equation for standard Gibbs energy change.

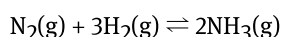
Equation	Symbols	Units
$\Delta G^\ominus = -RT \ln K$	$\Delta G^\ominus$ is the standard Gibbs energy change for the reaction	joules per mole (J mol ⁻¹)
	R is the universal gas constant (8.31 J K ⁻¹ mol ⁻¹)	joules per kelvin mole (J K ⁻¹ mol ⁻¹)
	T = the absolute temperature	kelvin (K)
	$\ln K$ = the natural logarithm of the equilibrium constant	

Note that R has the units J K⁻¹ mol⁻¹, so $\Delta G^\ominus$ in this equation has units of J mol⁻¹. To convert to kJ mol⁻¹, divide by 1000.

From the equation above, you can notice that ΔG is directly proportional to the natural logarithm of K . This means that when a reaction favours the formation of products at equilibrium (large K), the forward reaction is more spontaneous (ΔG negative). When the reaction has a low value of K , it means that the reaction favours the formation of reactants at equilibrium, and the ΔG change for the reaction is less negative (the forward reaction is less spontaneous) or even positive (the backward reaction is spontaneous).

Worked example 3

Consider again the reaction used to produce ammonia.



The standard Gibbs energy, $\Delta G^\ominus$, is -32.7 kJ mol⁻¹ and $T = 25^\circ\text{C}$.

Determine the equilibrium constant, K .

Solution steps	Calculations
Step 1: Write out the information provided in the question.	$\Delta G^\ominus$, is -32.7 kJ mol ⁻¹ $T = 25^\circ\text{C}$
Step 2: Convert any values that need changing.	$T = 25 + 273$ $= 298 \text{ K}$
Step 3: Write out the equation.	$\Delta G^\ominus = -RT \ln K$

Solution steps	Calculations
Step 4 Substitute the values given.	$\ln K = -\frac{\Delta G^\ominus}{RT}$ $= -\frac{-32.7}{\left(\frac{8.31}{1000}\right)(298)}$ $= 13.2$ $K = e^{13.2}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 5.4 \times 10^5 \text{ (2 s.f.)}$ Note that as $K \gg 1$, the forward reaction is favoured and, at equilibrium, the concentrations of the products are much greater than those of the reactants.

Study skills

The equations that relate Gibbs energy, ΔG , with the extent of the reaction (Q or K) for a reversible reaction are given in the DP chemistry data booklet in section 1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>).

$$\Delta G = \Delta G^\ominus + RT \ln Q$$

$$\Delta G^\ominus = -RT \ln K$$

As a reversible reaction proceeds, the composition of the reaction mixture is constantly changing, as is the Gibbs energy. The position of equilibrium corresponds to a maximum value of entropy (S) and a minimum value of Gibbs energy change (ΔG). Once this point has been reached, the reaction will not proceed any further – the rates of the forward reaction and the reverse reaction are now equal. Which means the reaction quotient, Q , is equal to the equilibrium constant, K , ($Q = K$).

By substituting the first equation into the second we get:

$$\Delta G = -RT \ln K + RT \ln Q$$

Therefore:

$$\Delta G = RT \left(\ln \frac{Q}{K} \right)$$

Recall, that at equilibrium, $Q = K$, therefore, $\frac{Q}{K} = 1$. The natural log ($\ln$) of 1 = 0, so:

$$\Delta G = RT(\ln 1) = 0$$

From this, we can see that at equilibrium, when $Q = K$, the ΔG for a reaction is zero.

Note that the standard change in Gibbs energy ($\Delta G^\ominus$) only has a value of zero when the equilibrium constant, K , is equal to 1:

$$\Delta G^\ominus = -RT \ln(1) = 0$$

Figure 3 shows that when the equilibrium constant, K , has a value of 1, the value of the $\Delta G^\ominus$ is zero. For such a reaction, this means that neither the products nor reactants are favoured at equilibrium.

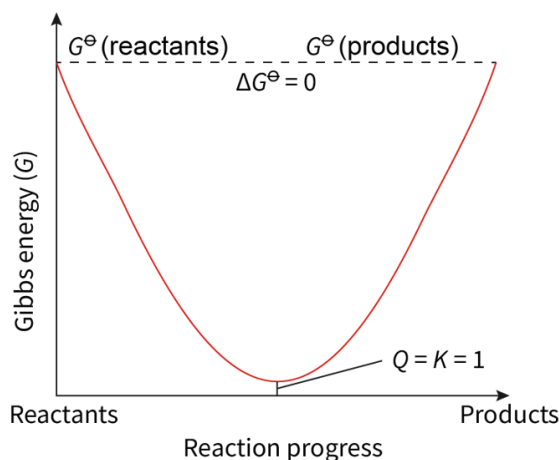


Figure 3. The standard change in Gibbs energy ($\Delta G^\ominus$) only has a value of zero when the equilibrium constant, K , is equal to 1

 More information for figure 3

The graph is titled with measures of Gibbs energy (G) against the reaction progress, depicted with a curve. The X-axis represents 'Reaction progress', starting from 'Reactants' on the left to 'Products' on the right. The Y-axis is labeled as 'Gibbs energy (G)'. The curve displays an initial high Gibbs energy at the reactants, decreasing to a minimum point before ascending again to the product side, illustrating a concave-upward U-shaped curve.

Key points of the graph include labels: $G^\ominus$ (reactants) and $G^\ominus$ (products), with a dashed line running parallel to the X-axis at these points. The text ' $\Delta G^\ominus = 0$ ' is positioned between the dashed lines for reactants and products, suggesting that the standard change in Gibbs energy is zero at equilibrium. An annotation on the curve highlights ' $Q = K = 1$,' indicating the equilibrium condition where the reaction is perfectly balanced between products and reactants.

[Generated by AI]

Gibbs energy and spontaneity of a reversible reaction

Given the equations studied above, you can use the values of standard Gibbs energy change, $\Delta G^\ominus$, and the equilibrium constant, K , to determine if the forward reaction of a reversible reaction is spontaneous or not.

Compare the values in **Table 3** that shows K and $\Delta G^\ominus$ for various chemical reactions.

Table 3. Values of $\Delta G^\ominus$ and K for selected reactions at 298 K.

Reaction	$\Delta G^\ominus$ (kJ mol ⁻¹)	K
$2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$	+142	1.4×10^{-10}
$\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$	+79.9	1.0×10^{-14}
$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$	-32.9	5.8×10^5
$\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu}(\text{s})$	-211	1.4×10^{37}

You might notice that the more negative values of standard Gibbs energy change, $\Delta G^\ominus$, mean larger values of the equilibrium constant, K . This means that the more spontaneous a reaction is, the further the position of equilibrium lies to the right (i.e. the forward reaction (products) is favoured).

Use **Interactive 3** to summarise the relationship between the equilibrium constant, K , and the standard Gibbs energy change, $\Delta G^\ominus$.

$\Delta G^\ominus$	$\ln K$	K	Position of equilibrium
Negative			
	Zero		
		< 1	

Negative

Positive

Zero

< 1

= 1

> 1

Lies to the left – favouring the reactants

Lies to the right – favouring the products

Equal proportions of reactants and products

☒ Check

Interactive 3. Summarise the relationship between K , and $\Delta G^\ominus$.

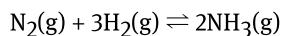
When the position of the equilibrium lies to the right on a reversible reaction, this means that the forward reaction is spontaneous. For example, for the reactions listed in the interactive:

- The reaction between zinc and aqueous copper ions has a large negative value for the $\Delta G^\ominus$ and a very large value for the equilibrium constant, K . From this, we can deduce that the reaction goes almost to completion, with the equilibrium mixture containing mostly products.

- The decomposition of sulfur trioxide, SO_3 , has a large positive value for the $\Delta G^\ominus$ and a very small value for the equilibrium constant, K . From this, we can deduce that, at 298K, the forward reaction barely proceeds, and the equilibrium mixture contains mostly reactants.

Worked example 4

Consider the reaction to produce ammonia.



You calculated the value of Gibbs energy, $\Delta G^\ominus = -72.5 \text{ kJ mol}^{-1}$ and the equilibrium constant, $K = 5.4 \times 10^5$.

Determine if the reaction favours the products or reactants.

Given that $\Delta G < 0$ and $K > 1$, the forward reaction is spontaneous and so the products are favoured at 25°C (298 K).

Figure 4 shows how the Gibbs energy changes for spontaneous and non-spontaneous reactions. For both, the Gibbs energy decreases until it reaches a minimum value, when the reaction is at equilibrium. This point also corresponds to the point of maximum entropy for the reaction, because mixtures have higher entropy than pure substances.

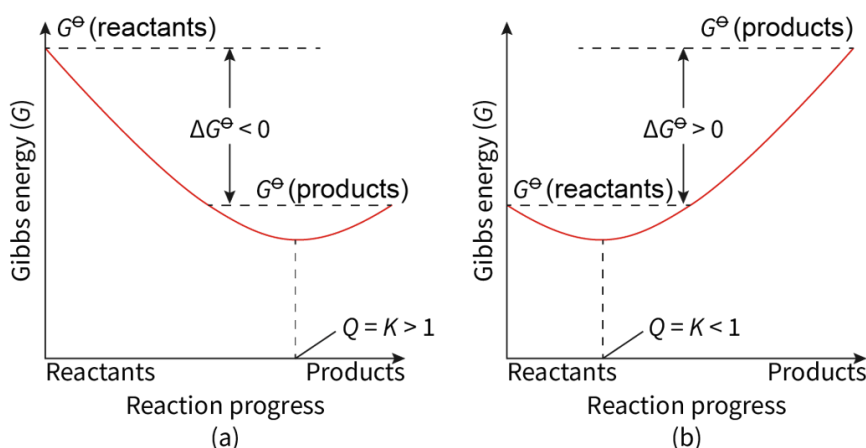


Figure 4. Plot of Gibbs energy versus reaction progress for spontaneous (a) and non-spontaneous (b) reactions.

 More information for figure 4

The image contains two graphs depicting the relationship between Gibbs energy and reaction progress for spontaneous and non-spontaneous reactions.

On the left (a): The graph shows a curve that starts high at the Reactants side and decreases towards the Products side, indicating a spontaneous reaction. The X-axis represents Reaction progress with labels for Reactants on the left and Products on the right. The Y-axis represents Gibbs energy (G), with the notation $G^\ominus$ for reactants and products. The curve dips to a minimum point at equilibrium, marked with $G^\ominus(\text{products})$ lower than $G^\ominus(\text{reactants})$. $\Delta G^\ominus$ is less than 0, and Q equals K which is greater than 1.

On the right (b): This graph illustrates a non-spontaneous reaction where the curve rises as it moves from Reactants to Products. The X-axis also represents Reaction progress, and the Y-axis is the Gibbs energy (G). The energy at G° (products) is higher than G° (reactants), with ΔG° greater than 0, and Q equals K , which is less than 1. The minimum energy point is positioned closer to Reactants, signifying higher reactants at equilibrium.

[Generated by AI]

For a spontaneous reaction, the position of equilibrium lies towards the products side. This is shown by the position of the minimum value of the Gibbs energy, which is closer to the products side than the reactants side, i.e. there is a higher concentration of products than reactants at equilibrium, with a K value greater than 1.

For a non-spontaneous reaction, this point on the graph lies more towards the reactants side than the products side. This means that at equilibrium, there is a greater concentration of reactants than products in the reaction mixture, with a K value of less than 1.

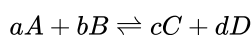
Making connections

Subtopic R2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>) What is the likely composition of an equilibrium mixture when ΔG° is positive?

For a reversible reaction, you can analyse the relationship between the Gibbs energy and extent of the reaction through the equation:

$$\Delta G^\circ = -RT \ln K$$

When ΔG is positive, this means that the forward reaction is not spontaneous under the given conditions and that energy must be supplied to drive the reaction forward. In this case, $K < 1$. In a hypothetical reversible reaction:



the equilibrium constant is written as:

$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

This means that for a reaction where $K \ll 1$, the concentration of products is much lower than the concentration of reactants at equilibrium.

Try the activity in the next section to investigate the effect of temperature on reaching equilibrium.